mental_effort_task_difficulty_HRv

October 11, 2026

1 Multilevel Analysis for Mental Effort, Task Difficulty, Perceived Performance Perception, and Team Gameplay Score based on Cardiac Measures

by Elvis Ortega-Ochoa

[1]:
```
import sys
import subprocess

subprocess.check_call([sys.executable, "-m", "pip", "install", "--upgrade",␣
 ↪"pip", "ipywidgets", "jupyterlab_widgets"])
subprocess.check_call([
    sys.executable, "-m", "pip", "install",
    "pymer4",
    "rpy2",
    "polars",
    "great-tables",
    "joblib",
    "scikit-learn",
    "pyarrow",
    "bambi",
    "arviz",
    "matplotlib",
    "pandas",
    "numpy",
    "statsmodels",
    "openpyxl",
])
```

```
Requirement already satisfied: pip in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(26.2)
Collecting pip
  Using cached pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
Requirement already satisfied: ipywidgets in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(8.1.8)
Collecting ipywidgets
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  Using cached ipywidgets-8.1.9-py3-none-any.whl.metadata (2.4 kB)
Requirement already satisfied: jupyterlab_widgets in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(3.0.16)
Collecting jupyterlab_widgets
  Using cached jupyterlab_widgets-3.0.17-py3-none-any.whl.metadata (20 kB)
Requirement already satisfied: comm>=0.1.3 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipywidgets) (0.2.2)
Requirement already satisfied: ipython>=6.1.0 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipywidgets) (9.2.0)
Requirement already satisfied: traitlets>=4.3.1 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipywidgets) (5.14.3)
Collecting widgetsnbextension~=4.0.16 (from ipywidgets)
  Using cached widgetsnbextension-4.0.16-py3-none-any.whl.metadata (1.6 kB)
Requirement already satisfied: decorator in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (5.2.1)
Requirement already satisfied: ipython-pygments-lexers in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (1.1.1)
Requirement already satisfied: jedi>=0.16 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (0.19.2)
Requirement already satisfied: matplotlib-inline in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (0.1.7)
Requirement already satisfied: pexpect>4.3 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (3.0.51)
Requirement already satisfied: pygments>=2.4.0 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (2.19.1)
Requirement already satisfied: stack_data in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (0.6.3)
Requirement already satisfied: wcwidth in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
prompt_toolkit<3.1.0,>=3.0.41->ipython>=6.1.0->ipywidgets) (0.2.13)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
jedi>=0.16->ipython>=6.1.0->ipywidgets) (0.8.4)
Requirement already satisfied: ptyprocess>=0.5 in
```

```
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
pexpect>4.3->ipython>=6.1.0->ipywidgets) (0.7.0)
Requirement already satisfied: executing>=1.2.0 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
stack_data->ipython>=6.1.0->ipywidgets) (2.2.0)
Requirement already satisfied: asttokens>=2.1.0 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
stack_data->ipython>=6.1.0->ipywidgets) (3.0.0)
Requirement already satisfied: pure-eval in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
stack_data->ipython>=6.1.0->ipywidgets) (0.2.3)
Using cached pip-26.2.1-py3-none-any.whl (1.8 MB)
Using cached ipywidgets-8.1.9-py3-none-any.whl (140 kB)
Using cached jupyterlab_widgets-3.0.17-py3-none-any.whl (217 kB)
Using cached widgetsnbextension-4.0.16-py3-none-any.whl (2.2 MB)
Installing collected packages: widgetsnbextension, pip, jupyterlab_widgets,
ipywidgets
  Attempting uninstall: widgetsnbextension
    Found existing installation: widgetsnbextension 4.0.15
    Uninstalling widgetsnbextension-4.0.15:
      Successfully uninstalled widgetsnbextension-4.0.15
  Attempting uninstall: pip
    Found existing installation: pip 26.2
    Uninstalling pip-26.2:
      Successfully uninstalled pip-26.2
  Attempting uninstall: jupyterlab_widgets
1/4 [pip]
    Found existing installation: jupyterlab_widgets 3.0.16
1/4 [pip]
    Uninstalling jupyterlab_widgets-3.0.16:
1/4 [pip]
      Successfully uninstalled jupyterlab_widgets-3.0.16
1/4 [pip]
  Attempting uninstall: ipywidgets
1/4 [pip]
    Found existing installation: ipywidgets 8.1.8
1/4 [pip]
    Uninstalling ipywidgets-8.1.8:
1/4 [pip]
      Successfully uninstalled ipywidgets-8.1.8
1/4 [pip]
                          4/4
[ipywidgets]4 [ipywidgets]
Successfully installed ipywidgets-8.1.9 jupyterlab_widgets-3.0.17
pip-26.2.1 widgetsnbextension-4.0.16
Requirement already satisfied: pymer4 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(0.9.2)
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```
Requirement already satisfied: rpy2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(3.6.7)
Requirement already satisfied: polars in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(1.43.2)
Requirement already satisfied: great-tables in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(0.23.0)
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(1.5.3)
Requirement already satisfied: scikit-learn in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(1.9.0)
Requirement already satisfied: pyarrow in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(25.0.0)
Requirement already satisfied: bambi in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(0.19.0)
Requirement already satisfied: arviz in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(1.2.0)
Requirement already satisfied: matplotlib in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(3.10.3)
Requirement already satisfied: pandas in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(3.0.5)
Requirement already satisfied: numpy in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(2.2.6)
Requirement already satisfied: statsmodels in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(0.14.6)
Requirement already satisfied: openpyxl in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(3.1.5)
Requirement already satisfied: rpy2-rinterface>=3.6.6 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from rpy2) (3.6.6)
Requirement already satisfied: rpy2-robjects>=3.6.5 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from rpy2) (3.6.5)
Requirement already satisfied: polars-runtime-32==1.43.2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from polars) (1.43.2)
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Requirement already satisfied: multimark>=0.1.3 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (0.2.0)
Requirement already satisfied: faicons>=0.2.2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (0.2.2)
Requirement already satisfied: htmltools>=0.4.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (0.7.0)
Requirement already satisfied: importlib-metadata in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (9.0.0)
Requirement already satisfied: nokap>=0.1.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (0.1.0)
Requirement already satisfied: typing_extensions>=3.10.0.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (4.16.0)
Requirement already satisfied: Babel>=2.13.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (2.18.0)
Requirement already satisfied: importlib-resources in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (7.1.0)
Requirement already satisfied: scipy>=1.10.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from scikit-learn) (1.18.0)
Requirement already satisfied: narwhals>=2.0.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from scikit-learn) (2.24.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from scikit-learn) (3.6.0)
Requirement already satisfied: arviz_plots<2.0.0,>=1.0.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (1.2.0)
Requirement already satisfied: formulae<0.7.0,>=0.6.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (0.6.2)
Requirement already satisfied: graphviz in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (0.21)
Requirement already satisfied: pytensor<4.0.0,>=3.0.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (3.2.4)
Requirement already satisfied: pymc<7.0.0,>=6.0.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (6.2.0)
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Requirement already satisfied: seaborn<0.14,>=0.13.2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (0.13.2)
Requirement already satisfied: sparse<0.18,>=0.17.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (0.17.0)
Requirement already satisfied: arviz-base<1.3,>=1.2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from arviz_plots<2.0.0,>=1.0.0->bambi) (1.2.0)
Requirement already satisfied: arviz-stats<1.3,>=1.2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from arviz-stats[xarray]<1.3,>=1.2->arviz_plots<2.0.0,>=1.0.0->bambi) (1.2.0)
Requirement already satisfied: xarray>=2024.11.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from arviz-base<1.3,>=1.2->arviz_plots<2.0.0,>=1.0.0->bambi) (2026.7.0)
Requirement already satisfied: lazy_loader>=0.4 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from arviz-base<1.3,>=1.2->arviz_plots<2.0.0,>=1.0.0->bambi) (0.5)
Requirement already satisfied: xarray-einstats in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from arviz-stats[xarray]<1.3,>=1.2->arviz_plots<2.0.0,>=1.0.0->bambi) (0.11.0)
Requirement already satisfied: packaging in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
formulae<0.7.0,>=0.6.0->bambi) (25.0)
Requirement already satisfied: cachetools<7,>=4.2.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pymc<7.0.0,>=6.0.0->bambi) (6.2.6)
Requirement already satisfied: cloudpickle in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pymc<7.0.0,>=6.0.0->bambi) (3.1.2)
Requirement already satisfied: rich>=13.7.1 in
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(from pymc<7.0.0,>=6.0.0->bambi) (15.0.0)
Requirement already satisfied: setuptools>=59.0.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pytensor<4.0.0,>=3.0.0->bambi) (83.0.0)
Requirement already satisfied: numba<=0.66.0,>=0.58 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pytensor<4.0.0,>=3.0.0->bambi) (0.66.0)
Requirement already satisfied: filelock>=3.15 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pytensor<4.0.0,>=3.0.0->bambi) (3.32.2)
Requirement already satisfied: llvmlite<0.49,>=0.48.0dev0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from numba<=0.66.0,>=0.58->pytensor<4.0.0,>=3.0.0->bambi) (0.48.0)
Requirement already satisfied: contourpy>=1.0.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (1.3.2)
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Requirement already satisfied: cycler>=0.10 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (4.58.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (1.4.8)
Requirement already satisfied: pillow>=8 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (11.2.1)
Requirement already satisfied: pyparsing>=2.3.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
matplotlib) (2.9.0.post0)
Requirement already satisfied: patsy>=0.5.6 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from statsmodels) (1.0.2)
Requirement already satisfied: et-xmlfile in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from openpyxl) (2.0.0)
Requirement already satisfied: cffi>=1.15 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from multimark>=0.1.3->great-tables) (2.1.0)
Requirement already satisfied: click>=8.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from multimark>=0.1.3->great-tables) (8.4.2)
Requirement already satisfied: pycparser in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from cffi>=1.15->multimark>=0.1.3->great-tables) (3.0)
Requirement already satisfied: websockets>=12.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from nokap>=0.1.0->great-tables) (17.0.1)
Requirement already satisfied: six>=1.5 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from python-
dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from rich>=13.7.1->pymc<7.0.0,>=6.0.0->bambi) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
rich>=13.7.1->pymc<7.0.0,>=6.0.0->bambi) (2.19.1)
Requirement already satisfied: mdurl~=0.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from markdown-it-py>=2.2.0->rich>=13.7.1->pymc<7.0.0,>=6.0.0->bambi) (0.1.2)
```

```
Requirement already satisfied: jinja2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from rpy2-robjects>=3.6.5->rpy2) (3.1.6)
Requirement already satisfied: tzlocal in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from rpy2-robjects>=3.6.5->rpy2) (5.4.4)
Requirement already satisfied: zipp>=3.20 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from importlib-metadata->great-tables) (4.1.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from jinja2->rpy2-robjects>=3.6.5->rpy2) (3.0.3)
```

[1]:
```
0
```

Packages

[3]:
```
import pandas as pd
import numpy as np
import bambi as bmb
import arviz as az
import matplotlib.pyplot as plt
import statsmodels.formula.api as smf
import matplotlib.patches as patches
import statsmodels.api as sm
import pymer4.models as pm
import polars as pl
import ipywidgets as widgets
from pymer4.models import lmer as Lmer
```

Loading data

[5]:
```
# Reading files and printing columns
hrv = pd.read_excel("../HRV_data_v3.xlsx", keep_default_na=False)
dv = pd.read_excel("../
 ↪CONT_dependent_variables_confidence_mental_effort_task_difficulty_E.xlsx",␣
 ↪keep_default_na=False)

# Make a clean copy
hrv = hrv.copy()
me_td = dv.copy()

# Extract participantID and sensorID from hrvID
hrv["participantID"] = hrv["hrvID"].str.extract(r"^([^.]+)\.", expand=False)
hrv["sensorID"] = hrv["hrvID"].str.extract(r"\.([A-Z])-PPG", expand=False) * 2

# Merge the dataframes on groupID, participantID, and sensorID
dataset = hrv.merge(
```

```
    me_td,
    on=["groupID", "participantID", "sensorID"],
    how="inner"
)

# Rename columns for clarity
dataset = dataset.rename(columns={
    "ME1": "me",
    "TD1": "td",
    "PP1": "pp"
})
```

Reverse team gameplay score

[6]:
```
# Build numeric uncertScore
dataset["uncertScore"] = pd.to_numeric(dataset["uncertScore"], errors="coerce")

# Reverse-coding, using the instrument's theoretical bounds, not the sample's
SCALE_MIN, SCALE_MAX = 1, 70  # instrument range
dataset["uncertScore"] = (SCALE_MIN + SCALE_MAX) - dataset["uncertScore"]
```

[7]:
```
# Finalize data types
dataset["groupID"] = dataset["groupID"].astype("category")
#dataset["cf"] = pd.to_numeric(dataset["cf"], errors="coerce")
dataset["me"] = pd.to_numeric(dataset["me"], errors="coerce")
dataset["td"] = pd.to_numeric(dataset["td"], errors="coerce")
dataset["pp"] = pd.to_numeric(dataset["pp"], errors="coerce")
```

[8]:
```
# Keep just variables needed
dat_model = dataset[["groupID", "participantID", "me", "td", "pp",␣
 ↪"uncertScore", "realMean_HR_bpm", "realRMSSD_ms"]].copy()
```

1.1 Mixed Linear Models

1.1.1 Within/Between Descomposition

[9]:
```
def add_within_between(df, raw_col, prefix):
    group_mean = df.groupby("groupID")[raw_col].transform("mean")
    within = df[raw_col] - group_mean
    between = group_mean - group_mean.mean()

    df[f"{prefix}_group_mean"] = group_mean
    df[f"{prefix}_within"] = within
    df[f"{prefix}_between"] = between

    return df

dat_model = add_within_between(dat_model, "realMean_HR_bpm", "hr")
```

```
dat_model = add_within_between(dat_model, "realRMSSD_ms", "rmssd")
dat_model = add_within_between(dat_model, "pp", "pp")
dat_model = add_within_between(dat_model, "me", "me")
dat_model = add_within_between(dat_model, "td", "td")
```

1.1.2 Standarize predictor variables

[10]:
```
dat_model["realMean_HR_bpm_z"] = (
    dat_model["realMean_HR_bpm"] - dat_model["realMean_HR_bpm"].mean()
) / dat_model["realMean_HR_bpm"].std()

dat_model["hr_within_z"] = (
    dat_model["hr_within"] - dat_model["hr_within"].mean()
) / dat_model["hr_within"].std()

dat_model["hr_between_z"] = (
    dat_model["hr_between"] - dat_model["hr_between"].mean()
) / dat_model["hr_between"].std()

dat_model["realRMSSD_ms_z"] = (
    dat_model["realRMSSD_ms"] - dat_model["realRMSSD_ms"].mean()
) / dat_model["realRMSSD_ms"].std()

dat_model["rmssd_within_z"] = (
    dat_model["rmssd_within"] - dat_model["rmssd_within"].mean()
) / dat_model["rmssd_within"].std()

dat_model["rmssd_between_z"] = (
    dat_model["rmssd_between"] - dat_model["rmssd_between"].mean()
) / dat_model["rmssd_between"].std()

dat_model["pp_z"] = (
    dat_model["pp"] - dat_model["pp"].mean()
) / dat_model["pp"].std()


dat_model["uncertScore_z"] = (
    dat_model["uncertScore"] - dat_model["uncertScore"].mean()
) / dat_model["uncertScore"].std()

dat_model["me_z"] = (
    dat_model["me"] - dat_model["me"].mean()
) / dat_model["me"].std()

dat_model["me_within_z"] = (
    dat_model["me_within"] - dat_model["me_within"].mean()
) / dat_model["me_within"].std()
```

```
dat_model["me_between_z"] = (
    dat_model["me_between"] - dat_model["me_between"].mean()
) / dat_model["me_between"].std()

dat_model["td_z"] = (
    dat_model["td"] - dat_model["td"].mean()
) / dat_model["td"].std()

dat_model["td_within_z"] = (
    dat_model["td_within"] - dat_model["td_within"].mean()
) / dat_model["td_within"].std()

dat_model["td_between_z"] = (
    dat_model["td_between"] - dat_model["td_between"].mean()
) / dat_model["td_between"].std()

dat_model.columns.tolist()

# Team-level dataset (one row per team, n = 29) for team gameplay score models
team_model = dat_model.drop_duplicates(subset="groupID").copy()
```

1.2 Exploratory Data Analysis

[11]:
```
cols_no_z = ["me", "td", "pp", "uncertScore", "realMean_HR_bpm", "realRMSSD_ms"]

# Labels to display
labels = {
    "me": "Mental effort",
    "me_z": "Mental effort Z-score",
    "td": "Task difficulty",
    "td_z": "Task difficulty Z-score",
    "pp": "Perceived indiv. performance",
    "pp_z": "Perceived indiv. performance Z-score",
    "uncertScore": "Team gameplay score",
    "uncertScore_z": "Team gameplay score Z-score",
    "realMean_HR_bpm": "Mean Heart Rate (bpm)",
    "realMean_HR_bpm_z": "Mean Heart Rate (bpm) Z-score",
    "hr_within_z": "Within-team Mean HR Z-score",
    "hr_between_z": "Between-team Mean HR Z-score",
    "realRMSSD_ms": "Root Mean Square of Successive Differences (RMSSD)",
    "realRMSSD_ms_z": "Root Mean Square of Successive Differences (RMSSD)␣
 ↪Z-score",
    "rmssd_within_z": "Within-team RMSSD Z-score",
    "rmssd_between_z": "Between-team RMSSD Z-score"
}
```

```
# Basic summary
print("Missing values:")
print(dat_model[cols_no_z].isna().sum(), "\n")

print("Descriptive statistics:")
display(dat_model[cols_no_z].describe().T)
```

```
Missing values:
me                 0
td                 0
pp                 0
uncertScore        0
realMean_HR_bpm    0
realRMSSD_ms       0
dtype: int64

Descriptive statistics:
```

	count	mean	std	min	25%	50%	\
me	87.0	6.402299	1.735135	3.0000	5.00000	6.0000	
td	87.0	6.390805	1.787270	2.0000	5.00000	7.0000	
pp	87.0	6.494253	1.764466	2.0000	5.00000	7.0000	
uncertScore	87.0	33.482759	7.603338	11.0000	30.00000	34.0000	
realMean_HR_bpm	87.0	-1.846626	11.973151	-70.3546	-3.14685	0.7487	
realRMSSD_ms	87.0	14.252398	28.079521	-41.8620	1.30885	9.9379	

	75%	max
me	7.00000	10.0000
td	8.00000	10.0000
pp	8.00000	10.0000
uncertScore	37.00000	46.0000
realMean_HR_bpm	2.59000	12.8109
realRMSSD_ms	20.87345	160.9357

[12]:
```
cols = ["me_z", "td_z", "pp_z", "uncertScore_z", "realMean_HR_bpm_z",␣
 ↪"realRMSSD_ms_z"]

# Basic summary
print("Missing values:")
print(dat_model[cols].isna().sum(), "\n")

print("Descriptive statistics:")
display(dat_model[cols].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols), figsize=(20, 8))

for i, col in enumerate(cols):
```

```
    axes[0, i].hist(dat_model[col], bins=12, edgecolor="black")
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

    axes[1, i].boxplot(dat_model[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

plt.tight_layout()
plt.show()

# Correlation matrix
corr = dat_model[cols].corr()

fig, ax = plt.subplots(figsize=(12, 9))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

display_labels = [labels[col] for col in cols]

ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)
ax.set_title("General Correlation Matrix")

threshold = 0.20
class MultiFocus:
    def __init__(self, items):
        self.items = set(items)
    def __eq__(self, other):
        return other in self.items

focus_var = MultiFocus(["me_z", "td_z"])

# Add correlation scores in each cell
for i in range(len(cols)):
    for j in range(len(cols)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols[i], cols[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
```

```
                                        linewidth=2.5, edgecolor="crimson",␣
↪facecolor="none")
            ax.add_patch(rect)

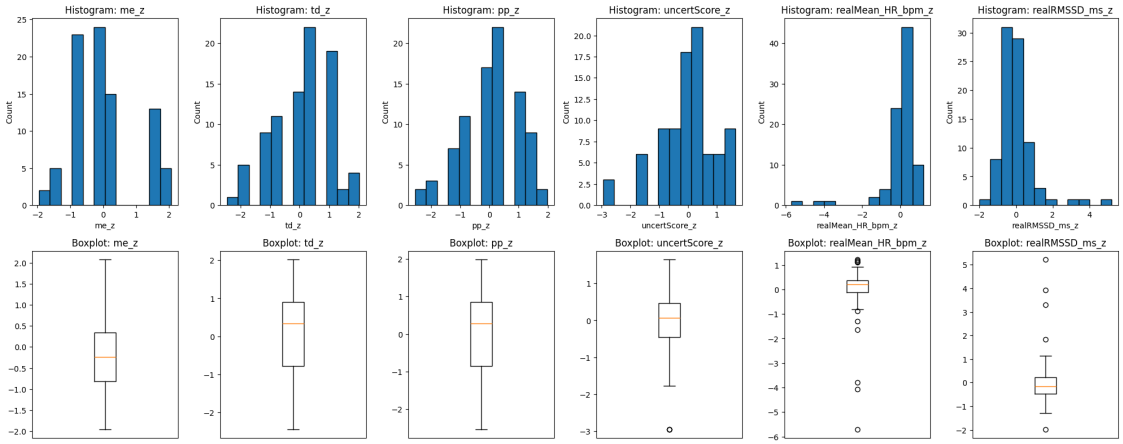
plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```

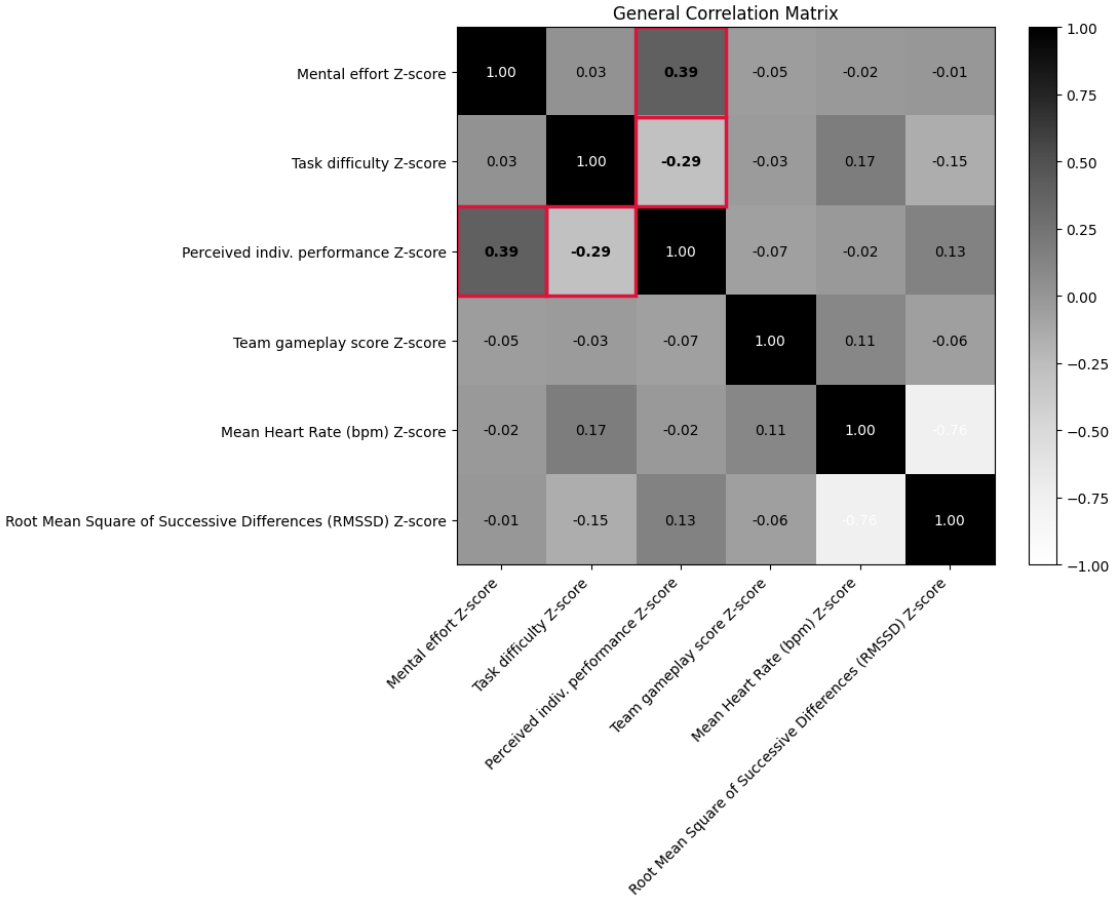
```
Missing values:
me_z                0
td_z                0
pp_z                0
uncertScore_z       0
realMean_HR_bpm_z   0
realRMSSD_ms_z      0
dtype: int64

Descriptive statistics:

                   count          mean  std       min       25%       50%  \
me_z                87.0 -1.065559e-16  1.0 -1.960827 -0.808179 -0.231855
td_z                87.0  1.528152e-16  1.0 -2.456711 -0.778173  0.340853
pp_z                87.0 -7.592905e-17  1.0 -2.547090 -0.846859  0.286629
uncertScore_z       87.0 -4.083579e-16  1.0 -2.956959 -0.458057  0.068028
realMean_HR_bpm_z   87.0 -3.605035e-17  1.0 -5.721800 -0.108595  0.216762
realRMSSD_ms_z      87.0 -1.276118e-17  1.0 -1.998410 -0.460960 -0.153653

                        75%       max
me_z                0.344470  2.073442
td_z                0.900365  2.019390
pp_z                0.853373  1.986861
uncertScore_z       0.462592  1.646282
realMean_HR_bpm_z   0.370548  1.224200
realRMSSD_ms_z      0.235796  5.223853
```





[13]:
```
eda_df = dat_model.copy()

cols_within_between = ["me_z", "td_z", "pp_z", "uncertScore_z", "hr_within_z",␣
 ↪"hr_between_z", "rmssd_within_z", "rmssd_between_z"]
cols_within = ["me_z", "td_z", "pp_z", "uncertScore_z", "hr_within_z",␣
 ↪"rmssd_within_z"]
cols_between = ["me_z", "td_z", "pp_z", "uncertScore_z", "hr_between_z",␣
 ↪"rmssd_between_z"]
```

Within-team Exploratory Data Analysis

[14]:
```
# Basic summary
print("Missing values:")
print(eda_df[cols_within].isna().sum(), "\n")

print("Descriptive statistics:")
```

```
display(eda_df[cols_within].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols_within), figsize=(20, 8))

for i, col in enumerate(cols_within):
    axes[0, i].hist(eda_df[col], bins=12, edgecolor="black")
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

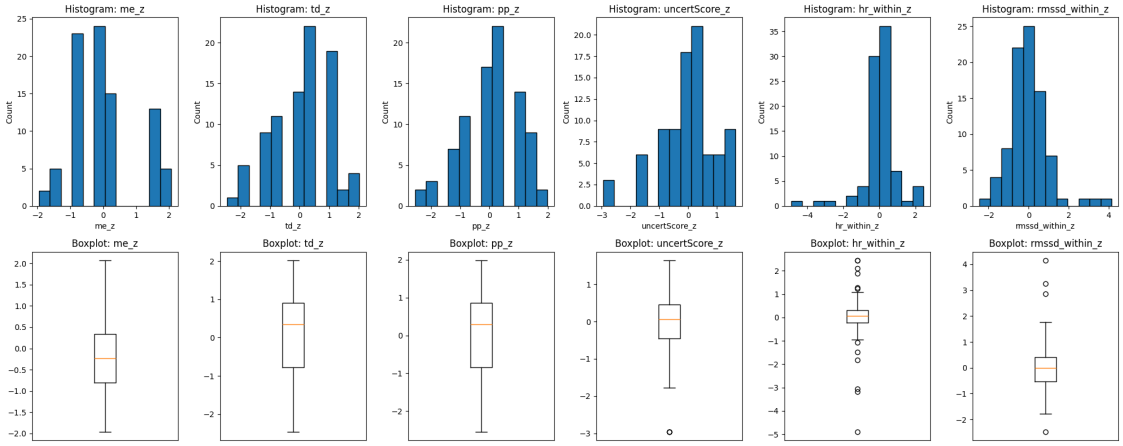
    axes[1, i].boxplot(eda_df[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

plt.tight_layout()
plt.show()
```

```
Missing values:
me_z             0
td_z             0
pp_z             0
uncertScore_z    0
hr_within_z      0
rmssd_within_z   0
dtype: int64

Descriptive statistics:
```

	count	mean	std	min	25%	50%	75%	max
me_z	87.0	-1.065559e-16	1.0	-1.960827	-0.808179	-0.231855	0.344470	2.073442
td_z	87.0	1.528152e-16	1.0	-2.456711	-0.778173	0.340853	0.900365	2.019390
pp_z	87.0	-7.592905e-17	1.0	-2.547090	-0.846859	0.286629	0.853373	1.986861
uncertScore_z	87.0	-4.083579e-16	1.0	-2.956959	-0.458057	0.068028	0.462592	1.646282
hr_within_z	87.0	-1.308021e-17	1.0	-4.896501	-0.220500	0.063628	0.325923	2.455934
rmssd_within_z	87.0	1.531342e-17	1.0	-2.473687	-0.524082	-0.012081	0.411390	4.157529



[15]:
```
# Correlation matrix
corr = eda_df[cols_within].corr()

fig, ax = plt.subplots(figsize=(12, 9))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

display_labels = [labels[col] for col in cols_within]

ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)

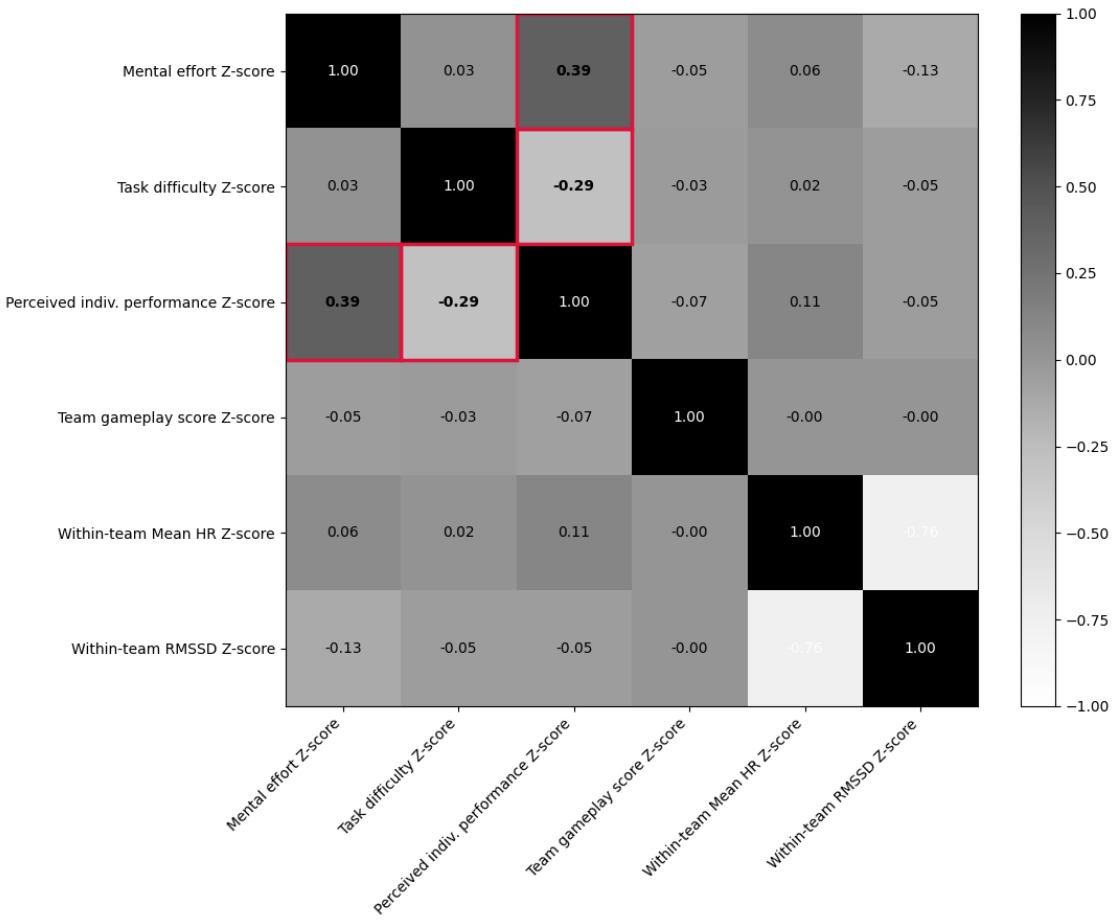
threshold = 0.20
class MultiFocus:
    def __init__(self, items):
        self.items = set(items)
    def __eq__(self, other):
        return other in self.items

focus_var = MultiFocus(["me_z", "td_z"])

for i in range(len(cols_within)):
    for j in range(len(cols_within)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_within[i], cols_within[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
```

```
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)

plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



Between-team Exploratory Data Analysis

[16]:
```
# Basic summary
print("Missing values:")
print(eda_df[cols_between].isna().sum(), "\n")

print("Descriptive statistics:")
display(eda_df[cols_between].describe().T)
```

```python
# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols_between), figsize=(20, 8))

for i, col in enumerate(cols_between):
    axes[0, i].hist(eda_df[col], bins=12, edgecolor="black")
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

    axes[1, i].boxplot(eda_df[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

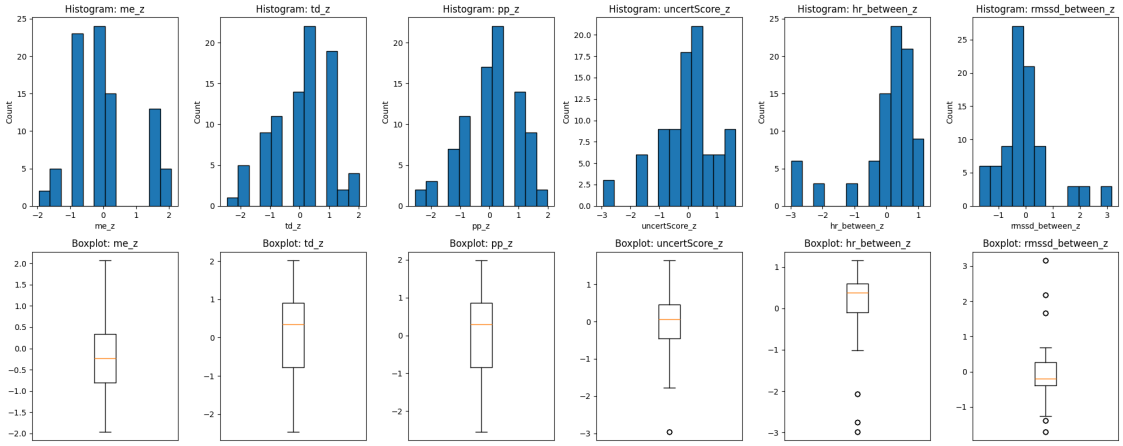
plt.tight_layout()
plt.show()
```

```
Missing values:
me_z              0
td_z              0
pp_z              0
uncertScore_z     0
hr_between_z      0
rmssd_between_z   0
dtype: int64

Descriptive statistics:

                 count          mean  std       min       25%       50%  \
me_z              87.0 -1.065559e-16  1.0 -1.960827 -0.808179 -0.231855
td_z              87.0  1.528152e-16  1.0 -2.456711 -0.778173  0.340853
pp_z              87.0 -7.592905e-17  1.0 -2.547090 -0.846859  0.286629
uncertScore_z     87.0 -4.083579e-16  1.0 -2.956959 -0.458057  0.068028
hr_between_z      87.0 -1.084701e-17  1.0 -2.981593 -0.100543  0.372596
rmssd_between_z   87.0 -6.380592e-18  1.0 -1.707508 -0.386674 -0.192074

                      75%       max
me_z             0.344470  2.073442
td_z             0.900365  2.019390
pp_z             0.853373  1.986861
uncertScore_z    0.462592  1.646282
hr_between_z     0.592498  1.162034
rmssd_between_z  0.270356  3.163902
```



[17]:
```
# Correlation matrix
corr = eda_df[cols_between].corr()

fig, ax = plt.subplots(figsize=(12, 9))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

display_labels = [labels[col] for col in cols_between]

ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)

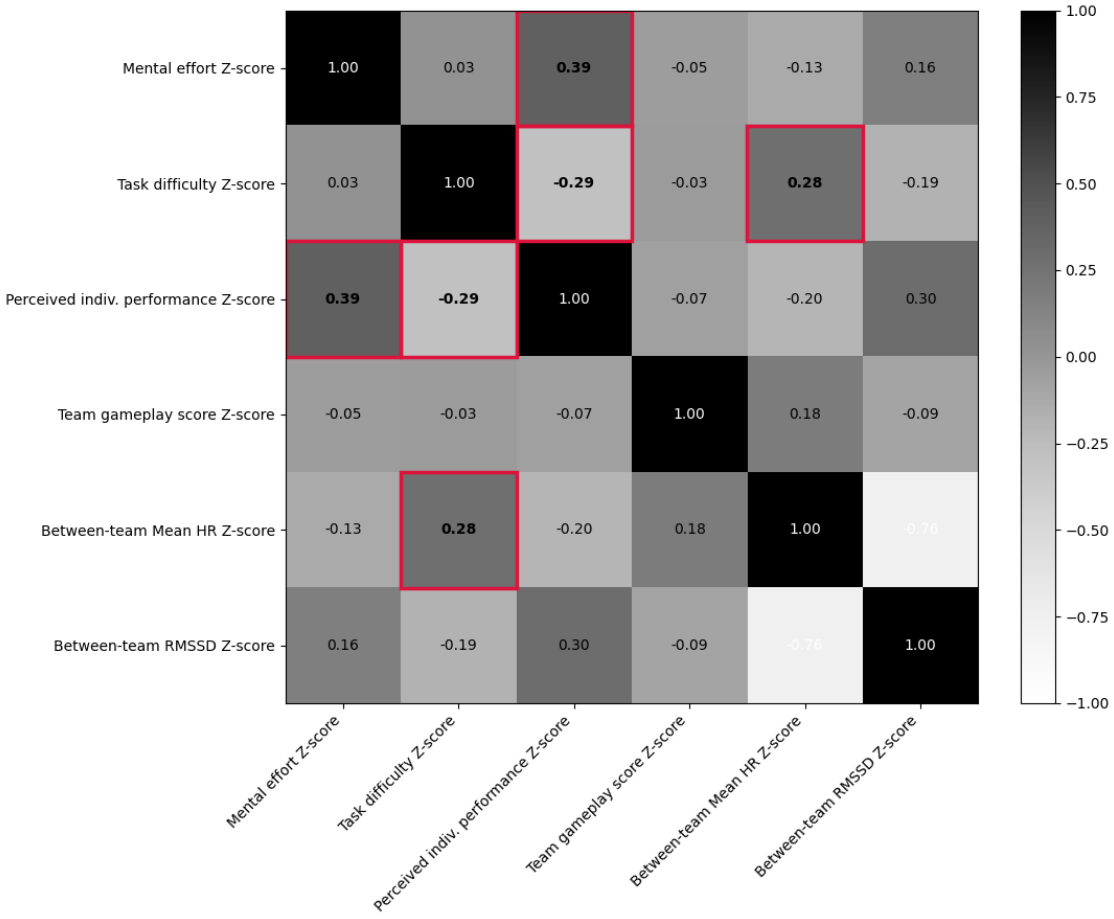
threshold = 0.20
class MultiFocus:
    def __init__(self, items):
        self.items = set(items)
    def __eq__(self, other):
        return other in self.items

focus_var = MultiFocus(["me_z", "td_z"])

for i in range(len(cols_between)):
    for j in range(len(cols_between)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_between[i], cols_between[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
```

```
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)

plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



[18]:
```
# Correlation matrix
corr = eda_df[cols_within_between].corr()

fig, ax = plt.subplots(figsize=(12, 9))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

display_labels = [labels[col] for col in cols_within_between]

ax.set_xticks(range(len(display_labels)))
```

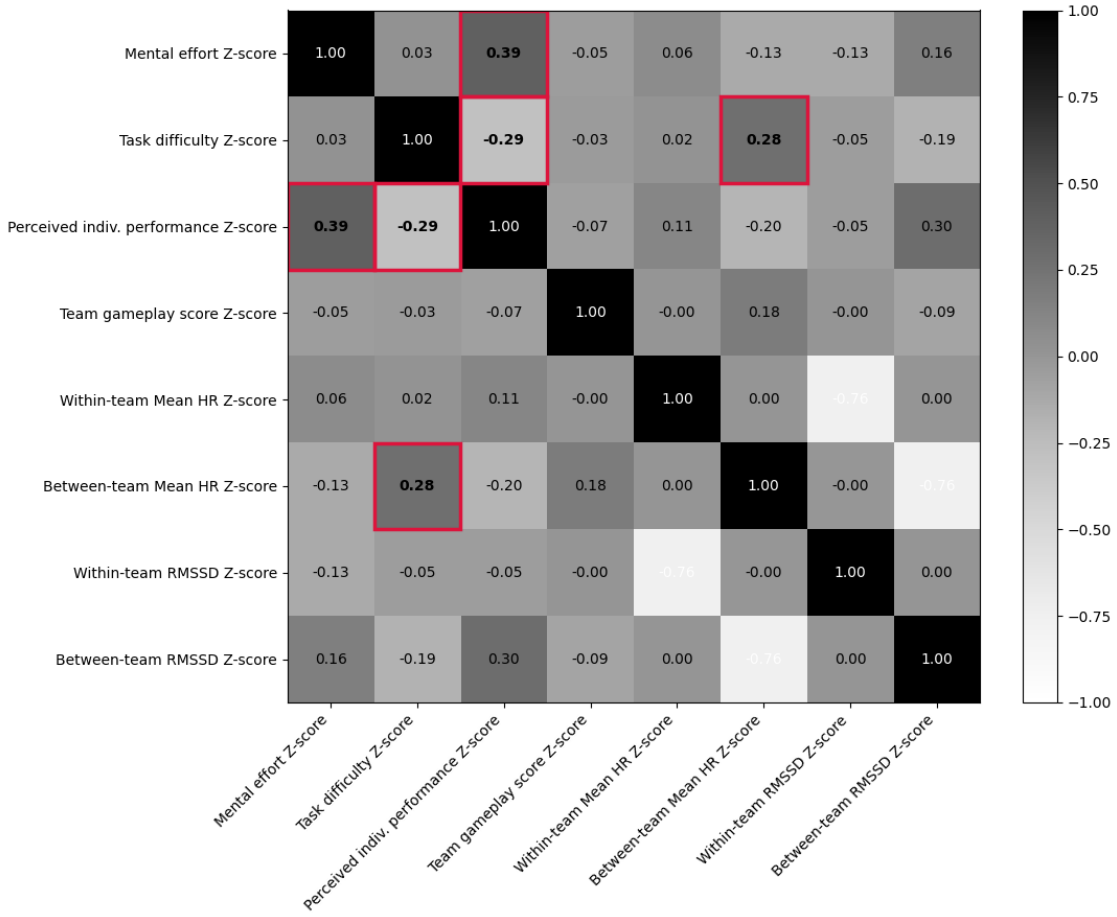
```
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)

threshold = 0.20
class MultiFocus:
    def __init__(self, items):
        self.items = set(items)
    def __eq__(self, other):
        return other in self.items

focus_var = MultiFocus(["me_z", "td_z"])

for i in range(len(cols_within_between)):
    for j in range(len(cols_within_between)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_within_between[i],␣
 ↪cols_within_between[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)

plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



[19]:
```
# Group ID as plain string for statsmodels MixedLM
dat_model["g"] = dat_model["groupID"].astype(str)
```

1.2.1 Task difficulty

1.2.2 Model 1: TD_Z ~ realMeanHR_bpm_z + (1|groupID)

1. Statsmodels (LME4)

[18]:
```
model1_sm = sm.MixedLM.from_formula(
    "td_z ~ realMean_HR_bpm_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result1_sm = model1_sm.fit(reml=True,method="nm", maxiter=5000)
print("Model 1 sm: TD_Z ~ realMean_HR_bpm_z + (1|groupID)")
print(result1_sm.summary())
```

```
Model 1 sm: TD_Z ~ realMean_HR_bpm_z + (1|groupID)
```

```
          Mixed Linear Model Regression Results
============================================================
Model:             MixedLM  Dependent Variable:  td_z
No. Observations:  87       Method:              REML
No. Groups:        29       Scale:               0.9808
Min. group size:   3        Log-Likelihood:      -124.2457
Max. group size:   3        Converged:           Yes
Mean group size:   3.0
------------------------------------------------------------
                  Coef. Std.Err.   z   P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept         0.000    0.106 0.000 1.000 -0.208  0.208
realMean_HR_bpm_z 0.175    0.113 1.555 0.120 -0.046  0.396
g Var             0.000    0.115
============================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

2. Bambi (Robustness testing)

[19]:
```
model1_bmb = bmb.Model(
    "td_z ~ realMean_HR_bpm_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1 = model1_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 1 bmb: TD_Z ~ realMean_HR_bpm_z + (1|groupID)")
az.summary(idata_model1, var_names=["realMean_HR_bpm_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, realMean_HR_bpm_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
```

```
total) took 2 seconds.
There was 1 divergence after tuning. Increase `target_accept` or reparameterize.
Model 1 bmb: TD_Z ~ realMean_HR_bpm_z + (1|groupID)
```

[19]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	\
realMean_HR_bpm_z	0.158	0.110	-0.050	0.364	0.001	0.001	
1|groupID_sigma	0.218	0.146	0.000	0.468	0.003	0.002	
sigma	0.981	0.081	0.835	1.134	0.001	0.001	

	ess_bulk	ess_tail	r_hat
realMean_HR_bpm_z	8729.0	4329.0	1.0
1|groupID_sigma	3022.0	3817.0	1.0
sigma	7221.0	3674.0	1.0

Within/Between Model 1: TD_Z ~ hr_between_z + hr_within_z + (1|groupID)

1. Statsmodels (LME4)

[20]:
```
model1_within_sm = sm.MixedLM.from_formula(
    "td_z ~ hr_within_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result1_within_sm = model1_within_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 1 within sm: TD_Z ~ hr_within_z + (1|groupID)")
print(result1_within_sm.summary())
```

```
Model 1 within sm: TD_Z ~ hr_within_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:            MixedLM Dependent Variable: td_z
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              0.9697
Min. group size:  3       Log-Likelihood:     -125.4800
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
---------------------------------------------------------
               Coef. Std.Err.   z   P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept      0.000    0.112 0.000 1.000 -0.220  0.220
hr_within_z    0.019    0.106 0.180 0.857 -0.189  0.227
g Var          0.042    0.121
=========================================================
```

[21]:
```
model1_between_sm = sm.MixedLM.from_formula(
    "td_z ~ hr_between_z",
```

```
    groups="g",
    re_formula="1",
    data=dat_model
)
result1_between_sm = model1_between_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 1 between sm: TD_Z ~ hr_between_z + (1|groupID)")
print(result1_between_sm.summary())
```

```
Model 1 between sm: TD_Z ~ hr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
============================================================
Model:             MixedLM Dependent Variable: td_z
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              0.9333
Min. group size:   3       Log-Likelihood:     -122.1372
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
------------------------------------------------------------
               Coef. Std.Err.   z   P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept      0.000    0.104 0.000 1.000 -0.203  0.203
hr_between_z   0.278    0.104 2.673 0.008  0.074  0.483
g Var          0.000    0.111
============================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

[22]:
```
model1_within_between_sm = sm.MixedLM.from_formula(
    "td_z ~ hr_within_z + hr_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result1_within_between_sm = model1_within_between_sm.fit(reml=True,␣
 ↪method="nm", maxiter=5000)
print("Model 1 within-between sm: TD_Z ~ hr_within_z + hr_between_z +␣
 ↪(1|groupID)")
print(result1_within_between_sm.summary())
```

```
Model 1 within-between sm: TD_Z ~ hr_within_z + hr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
============================================================
Model:             MixedLM Dependent Variable: td_z
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              0.9441
```

```
Min. group size:  3       Log-Likelihood:     -123.4604
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
-----------------------------------------------------
              Coef. Std.Err.   z   P>|z| [0.025 0.975]
-----------------------------------------------------
Intercept     0.000    0.104 0.000 1.000 -0.204  0.204
hr_within_z   0.019    0.105 0.182 0.855 -0.186  0.224
hr_between_z  0.278    0.105 2.658 0.008  0.073  0.484
g Var         0.000    0.114
=====================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

2. Bambi (Robustness testing)

[23]:
```
model1_within_bmb  = bmb.Model(
    "td_z ~ hr_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1_within_bmb = model1_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 1 within bmb: TD_Z ~ hr_within_z + (1|groupID)")
az.summary(idata_model1_within_bmb, var_names=["hr_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, hr_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 7 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 1 within bmb: TD_Z ~ hr_within_z + (1|groupID)
```

[23]:
```
                   mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
hr_within_z       0.020  0.108  -0.192    0.214      0.001    0.001    8316.0
1|groupID_sigma   0.249  0.158   0.000    0.507      0.003    0.002    2139.0
sigma             0.988  0.085   0.836    1.151      0.001    0.001    4278.0

                  ess_tail  r_hat
hr_within_z         4799.0    1.0
1|groupID_sigma     3532.0    1.0
sigma               2539.0    1.0
```

[24]:
```
model1_between_bmb  = bmb.Model(
    "td_z ~ hr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1_between_bmb = model1_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 1 between bmb: TD_Z ~ hr_between_z + (1|groupID)")
az.summary(idata_model1_between_bmb, var_names=["hr_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, hr_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 1 between bmb: TD_Z ~ hr_between_z + (1|groupID)
```

[24]:
```
                   mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
hr_between_z      0.278  0.114   0.064    0.495      0.001    0.001    9250.0
1|groupID_sigma   0.204  0.139   0.000    0.443      0.003    0.001    2316.0
sigma             0.960  0.078   0.813    1.107      0.001    0.001    8312.0

                  ess_tail  r_hat
```

```
hr_between_z       5678.0    1.0
1|groupID_sigma    2946.0    1.0
sigma              5714.0    1.0
```

[25]:
```
model1_within_between_bmb  = bmb.Model(
    "td_z ~ hr_within_z + hr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1_within_between_bmb = model1_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 1 within-between bmb: TD_Z ~ hr_within_z + hr_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model1_within_between_bmb, var_names=["hr_within_z",␣
 ↪"hr_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, hr_within_z, hr_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 1 within-between bmb: TD_Z ~ hr_within_z + hr_between_z + (1|groupID)
```

[25]:
```
                   mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
hr_within_z       0.019  0.105  -0.182    0.210      0.001    0.001   12993.0
hr_between_z      0.277  0.113   0.063    0.487      0.001    0.001    8427.0
1|groupID_sigma   0.197  0.136   0.000    0.434      0.003    0.001    2643.0
sigma             0.966  0.077   0.827    1.112      0.001    0.001    7607.0

                  ess_tail  r_hat
hr_within_z         6042.0    1.0
hr_between_z        6006.0    1.0
1|groupID_sigma     3068.0    1.0
sigma               5667.0    1.0
```

1.2.3 Model 2: TD_Z ~ realMeanRMSSD_bpm_z + (1|groupID)

1. Statsmodels (LME4)

[26]:
```
model2_sm = sm.MixedLM.from_formula(
    "td_z ~ realRMSSD_ms_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result2_sm = model2_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 2 sm: TD_Z ~ realRMSSD_ms_z + (1|groupID)")
print(result2_sm.summary())
```

```
Model 2 sm: TD_Z ~ realRMSSD_ms_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: td_z
No. Observations:   87      Method:             REML
No. Groups:         29      Scale:              0.9671
Min. group size:    3       Log-Likelihood:     -124.5453
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       0.000    0.109  0.000 1.000 -0.213  0.213
realRMSSD_ms_z -0.149    0.109 -1.370 0.171 -0.362  0.064
g Var           0.022    0.115
============================================================
```

2. Bambi (Robustness testing)

[27]:
```
model2_bmb = bmb.Model(
    "td_z ~ realRMSSD_ms_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2 = model2_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 bmb: TD_Z ~ realRMSSD_ms_z + (1|groupID)")
```

```
az.summary(idata_model2, var_names=["realRMSSD_ms_z", "1|groupID_sigma","sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, realRMSSD_ms_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There was 1 divergence after tuning. Increase `target_accept` or reparameterize.
```

```
Model 2 bmb: TD_Z ~ realRMSSD_ms_z + (1|groupID)
```

[27]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
realRMSSD_ms_z	-0.143	0.111	-0.349	0.070	0.001	0.001	10239.0	5496.0	1.0
1|groupID_sigma	0.232	0.150	0.000	0.480	0.003	0.002	2564.0	4206.0	1.0
sigma	0.980	0.081	0.830	1.135	0.001	0.001	6299.0	3904.0	1.0

Within/Between Model 2: TD_Z ~ rmssd_between_z + rmssd_within_z + (1|groupID)

1. Statsmodels (LME4)

[28]:
```
model2_within_sm = sm.MixedLM.from_formula(
    "td_z ~ rmssd_within_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result2_within_sm = model2_within_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 2 within sm: TD_Z ~ rmssd_within_z + (1|groupID)")
print(result2_within_sm.summary())
```

```
Model 2 within sm: TD_Z ~ rmssd_within_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: td_z
No. Observations:   87      Method:             REML
No. Groups:         29      Scale:              0.9661
Min. group size:    3       Log-Likelihood:     -125.3717
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
```

```
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       0.000    0.112  0.000 1.000 -0.220  0.220
rmssd_within_z -0.053    0.106 -0.499 0.617 -0.261  0.155
g Var           0.043    0.121
============================================================
```

[29]:
```
model2_between_sm = sm.MixedLM.from_formula(
    "td_z ~ rmssd_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result2_between_sm = model2_between_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 2 between sm: TD_Z ~ rmssd_between_z + (1|groupID)")
print(result2_between_sm.summary())
```

```
Model 2 between sm: TD_Z ~ rmssd_between_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM  Dependent Variable:  td_z
No. Observations:   87       Method:              REML
No. Groups:         29       Scale:               0.9536
Min. group size:    3        Log-Likelihood:      -124.0521
Max. group size:    3        Converged:           Yes
Mean group size:    3.0
------------------------------------------------------------
                Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept        0.000    0.109  0.000 1.000 -0.213  0.213
rmssd_between_z -0.186    0.109 -1.698 0.090 -0.400  0.029
g Var            0.025    0.116
============================================================
```

[30]:
```
model2_within_between_sm = sm.MixedLM.from_formula(
    "td_z ~ rmssd_within_z + rmssd_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result2_within_between_sm = model2_within_between_sm.fit(reml=True,␣
 ↪method="nm", maxiter=5000)
print("Model 2 within-between sm: TD_Z ~ rmssd_within_z + rmssd_between_z +␣
 ↪(1|groupID)")
print(result2_within_between_sm.summary())
```

```
Model 2 within-between sm: TD_Z ~ rmssd_within_z + rmssd_between_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:             MixedLM  Dependent Variable:  td_z
No. Observations:  87       Method:              REML
No. Groups:        29       Scale:               0.9661
Min. group size:   3        Log-Likelihood:      -125.2554
Max. group size:   3        Converged:           Yes
Mean group size:   3.0
------------------------------------------------------------
                Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept        0.000    0.109  0.000 1.000 -0.213  0.213
rmssd_within_z  -0.053    0.106 -0.499 0.617 -0.261  0.155
rmssd_between_z -0.186    0.109 -1.698 0.090 -0.400  0.029
g Var            0.020    0.115
============================================================
```

2. Bambi (Robustness testing)

[31]:
```
model2_within_bmb  = bmb.Model(
    "td_z ~ rmssd_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_within_bmb = model2_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 within bmb: TD_Z ~ rmssd_within_z + (1|groupID)")
az.summary(idata_model2_within_bmb, var_names=["rmssd_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, rmssd_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 14 divergences after tuning. Increase `target_accept` or
```

reparameterize.

Model 2 within bmb: TD_Z ~ rmssd_within_z + (1|groupID)

[31]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
rmssd_within_z	-0.053	0.108	-0.269	0.140	0.001	0.001	8667.0	4626.0	1.0
1|groupID_sigma	0.254	0.159	0.000	0.519	0.004	0.002	1990.0	3526.0	1.0
sigma	0.983	0.082	0.827	1.132	0.001	0.001	3746.0	3354.0	1.0

[32]:
```
model2_between_bmb  = bmb.Model(
    "td_z ~ rmssd_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_between_bmb = model2_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 between bmb: TD_Z ~ rmssd_between_z + (1|groupID)")
az.summary(idata_model2_between_bmb, var_names=["rmssd_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, rmssd_between_z, 1|groupID_sigma, 1|groupID_offset]

Output()

Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws total) took 4 seconds.

Model 2 between bmb: TD_Z ~ rmssd_between_z + (1|groupID)

[32]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
rmssd_between_z	-0.187	0.118	-0.411	0.029	0.001	0.001	8603.0	4626.0	1.0
1|groupID_sigma	0.230	0.151	0.000	0.487	0.003	0.001	2115.0	3526.0	1.0
sigma	0.975	0.079	0.828	1.121	0.001	0.001	7176.0	3354.0	1.0

```
                  ess_tail  r_hat
rmssd_between_z     6062.0    1.0
1|groupID_sigma     2389.0    1.0
sigma               6103.0    1.0
```

[33]:
```
model2_within_between_bmb  = bmb.Model(
    "td_z ~ rmssd_within_z + rmssd_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_within_between_bmb = model2_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 within-between bmb: TD_Z ~ rmssd_within_z + rmssd_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model2_within_between_bmb, var_names=["rmssd_within_z",␣
 ↪"rmssd_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, rmssd_within_z, rmssd_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.
```

```
Model 2 within-between bmb: TD_Z ~ rmssd_within_z + rmssd_between_z +
(1|groupID)
```

[33]:
```
                   mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
rmssd_within_z   -0.051  0.107  -0.256    0.144      0.001    0.001   12048.0
rmssd_between_z -0.186  0.118  -0.410    0.030      0.001    0.001    9621.0
1|groupID_sigma  0.230  0.152   0.000    0.494      0.003    0.001    1900.0
sigma            0.979  0.081   0.830    1.132      0.001    0.001    5913.0

                  ess_tail  r_hat
rmssd_within_z     5718.0    1.0
rmssd_between_z    5823.0    1.0
1|groupID_sigma    2410.0    1.0
```

```
sigma                5291.0     1.0
```

1.2.4 Mental effort

1.2.5 Model 7: ME_Z ~ realMeanHR_bpm_z + (1|groupID)

1. Statsmodels (LME4)

[34]:
```
model7_sm = sm.MixedLM.from_formula(
    "me_z ~ realMean_HR_bpm_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result7_sm = model7_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 7 sm: ME_Z ~ realMean_HR_bpm_z + (1|groupID)")
print(result7_sm.summary())
```

```
Model 7 sm: ME_Z ~ realMean_HR_bpm_z + (1|groupID)
           Mixed Linear Model Regression Results
============================================================
Model:               MixedLM  Dependent Variable:  me_z
No. Observations:    87       Method:              REML
No. Groups:          29       Scale:               1.0114
Min. group size:     3        Log-Likelihood:      -125.5497
Max. group size:     3        Converged:           Yes
Mean group size:     3.0
------------------------------------------------------------
                  Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept         -0.000    0.108 -0.000 1.000 -0.211  0.211
realMean_HR_bpm_z -0.020    0.111 -0.182 0.856 -0.238  0.197
g Var              0.000    0.120
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

2. Bambi (Robustness testing)

[35]:
```
model7_bmb = bmb.Model(
    "me_z ~ realMean_HR_bpm_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)
```

```
idata_model7 = model7_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 7 bmb: ME_Z ~ realMean_HR_bpm_z + (1|groupID)")
az.summary(idata_model7, var_names=["realMean_HR_bpm_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, realMean_HR_bpm_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 6 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 7 bmb: ME_Z ~ realMean_HR_bpm_z + (1|groupID)
```

[35]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	\
realMean_HR_bpm_z	-0.010	0.113	-0.217	0.214	0.001	0.001	
1|groupID_sigma	0.203	0.141	0.000	0.444	0.003	0.002	
sigma	1.000	0.082	0.848	1.148	0.001	0.001	

	ess_bulk	ess_tail	r_hat
realMean_HR_bpm_z	9216.0	5506.0	1.0
1|groupID_sigma	2812.0	3666.0	1.0
sigma	6882.0	4616.0	1.0

Within/Between Model 7: ME_Z ~ hr_between_z + hr_within_z + (1|groupID)

1. Statsmodels (LME4)

[36]:
```
model7_within_sm = sm.MixedLM.from_formula(
    "me_z ~ hr_within_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result7_within_sm = model7_within_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 7 within sm: ME_Z ~ hr_within_z + (1|groupID)")
print(result7_within_sm.summary())
```

```
Model 7 within sm: ME_Z ~ hr_within_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:            MixedLM Dependent Variable: me_z
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              1.0077
Min. group size:  3       Log-Likelihood:     -125.3946
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
---------------------------------------------------------
               Coef. Std.Err.   z   P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept      0.000    0.108 0.000 1.000 -0.211  0.211
hr_within_z    0.064    0.108 0.588 0.557 -0.149  0.276
g Var          0.000    0.117
=========================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

[37]:
```python
model7_between_sm = sm.MixedLM.from_formula(
    "me_z ~ hr_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result7_between_sm = model7_between_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 7 between sm: ME_Z ~ hr_between_z + (1|groupID)")
print(result7_between_sm.summary())
```

```
Model 7 between sm: ME_Z ~ hr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:            MixedLM Dependent Variable: me_z
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              0.9956
Min. group size:  3       Log-Likelihood:     -124.8823
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
---------------------------------------------------------
              Coef.  Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept    -0.000    0.107 -0.000 1.000 -0.210  0.210
hr_between_z -0.126    0.108 -1.175 0.240 -0.337  0.084
g Var         0.000    0.118
=========================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

[38]:
```
model7_within_between_sm = sm.MixedLM.from_formula(
    "me_z ~ hr_within_z + hr_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result7_within_between_sm = model7_within_between_sm.fit(reml=True,␣
 ↪method="nm", maxiter=5000)
print("Model 7 within-between sm: ME_Z ~ hr_within_z + hr_between_z +␣
 ↪(1|groupID)")
print(result7_within_between_sm.summary())
```

```
Model 7 within-between sm: ME_Z ~ hr_within_z + hr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: me_z
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              1.0033
Min. group size:   3       Log-Likelihood:     -126.0167
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
             Coef.  Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept    -0.000    0.107 -0.000 1.000 -0.210  0.210
hr_within_z   0.064    0.108  0.589 0.556 -0.148  0.275
hr_between_z -0.126    0.108 -1.170 0.242 -0.338  0.085
g Var         0.000    0.120
=========================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

2. Bambi (Robustness testing)

[39]:
```
model7_within_bmb  = bmb.Model(
    "me_z ~ hr_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
```

```
)

idata_model7_within_bmb = model7_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 7 within bmb: ME_Z ~ hr_within_z + (1|groupID)")
az.summary(idata_model7_within_bmb, var_names=["hr_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, hr_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 2 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 7 within bmb: ME_Z ~ hr_within_z + (1|groupID)
```

[39]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
hr_within_z	0.063	0.108	-0.136	0.265	0.001	0.002	8733.0	
1|groupID_sigma	0.201	0.139	0.000	0.442	0.003	0.002	2558.0	
sigma	0.996	0.081	0.845	1.150	0.001	0.001	6152.0	

	ess_tail	r_hat
hr_within_z	4514.0	1.0
1|groupID_sigma	3464.0	1.0
sigma	4129.0	1.0

[40]:
```
model7_between_bmb  = bmb.Model(
    "me_z ~ hr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model7_between_bmb = model7_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
```

```
    random_seed=123,
    target_accept=0.99
)

print("Model 7 between bmb: ME_Z ~ hr_between_z + (1|groupID)")
az.summary(idata_model7_between_bmb, var_names=["hr_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, hr_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 7 between bmb: ME_Z ~ hr_between_z + (1|groupID)
```

[40]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
hr_between_z	-0.127	0.117	-0.337	0.104	0.001	0.001	9436.0	
1|groupID_sigma	0.200	0.138	0.000	0.437	0.003	0.001	2590.0	
sigma	0.992	0.080	0.843	1.137	0.001	0.001	9123.0	

	ess_tail	r_hat
hr_between_z	5805.0	1.0
1|groupID_sigma	2686.0	1.0
sigma	5413.0	1.0

[41]:
```
model7_within_between_bmb  = bmb.Model(
    "me_z ~ hr_within_z + hr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model7_within_between_bmb = model7_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 7 within-between bmb: ME_Z ~ hr_within_z + hr_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model7_within_between_bmb, var_names=["hr_within_z",␣
 ↪"hr_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, hr_within_z, hr_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 7 within-between bmb: ME_Z ~ hr_within_z + hr_between_z + (1|groupID)
```

[41]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
hr_within_z	0.065	0.109	-0.145	0.261	0.001	0.001	12967.0	
hr_between_z	-0.126	0.119	-0.340	0.100	0.001	0.001	8571.0	
1|groupID_sigma	0.199	0.141	0.000	0.446	0.003	0.001	2596.0	
sigma	0.998	0.081	0.846	1.146	0.001	0.001	8201.0	

	ess_tail	r_hat
hr_within_z	6049.0	1.0
hr_between_z	5697.0	1.0
1|groupID_sigma	3029.0	1.0
sigma	5847.0	1.0

1.2.6 Model 8: ME_Z ~ realMeanRMSSD_bpm_z + (1|groupID)

1. Statsmodels (LME4)

[42]:
```
model8_sm = sm.MixedLM.from_formula(
    "me_z ~ realRMSSD_ms_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result8_sm = model8_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 8 sm: ME_Z ~ realRMSSD_ms_z + (1|groupID)")
print(result8_sm.summary())
```

```
Model 8 sm: ME_Z ~ realRMSSD_ms_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: me_z
No. Observations:   87      Method:             REML
No. Groups:         29      Scale:              1.0117
Min. group size:    3       Log-Likelihood:     -125.5632
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
```

```
------------------------------------------------------------
Intercept      -0.000    0.108 -0.000 1.000 -0.211  0.211
realRMSSD_ms_z -0.009    0.114 -0.082 0.934 -0.233  0.214
g Var           0.000    0.121
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

2. Bambi (Robustness testing)

[43]:
```
model8_bmb = bmb.Model(
    "me_z ~ realRMSSD_ms_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model8 = model8_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 8 bmb: ME_Z ~ realRMSSD_ms_z + (1|groupID)")
az.summary(idata_model8, var_names=["realRMSSD_ms_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, realRMSSD_ms_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There was 1 divergence after tuning. Increase `target_accept` or reparameterize.
```

```
Model 8 bmb: ME_Z ~ realRMSSD_ms_z + (1|groupID)
```

[43]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
realRMSSD_ms_z	-0.026	0.116	-0.239	0.190	0.001	0.002	10401.0	
1|groupID_sigma	0.209	0.144	0.000	0.459	0.003	0.002	3225.0	
sigma	0.998	0.081	0.843	1.145	0.001	0.001	6836.0	

```
                ess_tail  r_hat
realRMSSD_ms_z     4737.0    1.0
1|groupID_sigma    4078.0    1.0
sigma              3784.0    1.0
```

Within/Between Model 8: ME_Z ~ rmssd_between_z + rmssd_within_z + (1|groupID)

1. Statsmodels (LME4)

[44]:
```
model8_within_sm = sm.MixedLM.from_formula(
    "me_z ~ rmssd_within_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result8_within_sm = model8_within_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 8 within sm: ME_Z ~ rmssd_within_z + (1|groupID)")
print(result8_within_sm.summary())
```

```
Model 8 within sm: ME_Z ~ rmssd_within_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: me_z
No. Observations:   87      Method:             REML
No. Groups:         29      Scale:              0.9953
Min. group size:    3       Log-Likelihood:     -124.8678
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept      -0.000    0.107 -0.000 1.000 -0.210  0.210
rmssd_within_z -0.128    0.108 -1.187 0.235 -0.339  0.083
g Var           0.000    0.115
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

[45]:
```
model8_between_sm = sm.MixedLM.from_formula(
    "me_z ~ rmssd_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
```

```
)
result8_between_sm = model8_between_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 8 between sm: ME_Z ~ rmssd_between_z + (1|groupID)")
print(result8_between_sm.summary())
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

```
Model 8 between sm: ME_Z ~ rmssd_between_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM  Dependent Variable:  me_z
No. Observations:   87       Method:              REML
No. Groups:         29       Scale:               0.9866
Min. group size:    3        Log-Likelihood:      -124.4981
Max. group size:    3        Converged:           Yes
Mean group size:    3.0
------------------------------------------------------------
                Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       -0.000    0.106 -0.000 1.000 -0.209  0.209
rmssd_between_z  0.158    0.107  1.471 0.141 -0.052  0.368
g Var            0.000    0.120
============================================================
```

[46]:
```
model8_within_between_sm = sm.MixedLM.from_formula(
    "me_z ~ rmssd_within_z + rmssd_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result8_within_between_sm = model8_within_between_sm.fit(reml=True,␣
 ↪method="nm", maxiter=5000)
print("Model 8 within-between sm: ME_Z ~ rmssd_within_z + rmssd_between_z +␣
 ↪(1|groupID)")
print(result8_within_between_sm.summary())
```

```
Model 8 within-between sm: ME_Z ~ rmssd_within_z + rmssd_between_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM  Dependent Variable:  me_z
No. Observations:   87       Method:              REML
No. Groups:         29       Scale:               0.9817
Min. group size:    3        Log-Likelihood:      -125.1014
Max. group size:    3        Converged:           Yes
Mean group size:    3.0
```

```
------------------------------------------------------------
                Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       -0.000    0.106 -0.000 1.000 -0.208  0.208
rmssd_within_z  -0.128    0.107 -1.196 0.232 -0.337  0.082
rmssd_between_z  0.158    0.107  1.475 0.140 -0.052  0.367
g Var            0.000    0.120
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
  warnings.warn(msg, ConvergenceWarning)
```

2. Bambi (Robustness testing)

[47]:
```
model8_within_bmb  = bmb.Model(
    "me_z ~ rmssd_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model8_within_bmb = model8_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 8 within bmb: ME_Z ~ rmssd_within_z + (1|groupID)")
az.summary(idata_model8_within_bmb, var_names=["rmssd_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, rmssd_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 4 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 8 within bmb: ME_Z ~ rmssd_within_z + (1|groupID)
```

[47]:
```
                   mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
rmssd_within_z   -0.127  0.109  -0.332    0.082      0.001    0.002   10640.0
1|groupID_sigma   0.202  0.139   0.000    0.442      0.002    0.001    3017.0
sigma             0.991  0.079   0.842    1.137      0.001    0.001    7464.0

                  ess_tail  r_hat
rmssd_within_z      4786.0    1.0
1|groupID_sigma     4006.0    1.0
sigma               4230.0    1.0
```

[48]:
```
model8_between_bmb  = bmb.Model(
    "me_z ~ rmssd_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model8_between_bmb = model8_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 8 between bmb: ME_Z ~ rmssd_between_z + (1|groupID)")
az.summary(idata_model8_between_bmb, var_names=["rmssd_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, rmssd_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 8 between bmb: ME_Z ~ rmssd_between_z + (1|groupID)
```

[48]:
```
                   mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
rmssd_between_z   0.159  0.118  -0.056    0.391      0.001    0.001   10766.0
1|groupID_sigma   0.198  0.136   0.000    0.432      0.002    0.001    2783.0
sigma             0.988  0.079   0.842    1.136      0.001    0.001    9996.0

                  ess_tail  r_hat
rmssd_between_z     5998.0    1.0
1|groupID_sigma     2936.0    1.0
```

```
sigma              5900.0    1.0
```

[49]:
```
model8_within_between_bmb  = bmb.Model(
    "me_z ~ rmssd_within_z + rmssd_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model8_within_between_bmb = model8_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 8 within-between bmb: ME_Z ~ rmssd_within_z + rmssd_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model8_within_between_bmb, var_names=["rmssd_within_z",␣
 ↪"rmssd_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, rmssd_within_z, rmssd_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 8 within-between bmb: ME_Z ~ rmssd_within_z + rmssd_between_z +
(1|groupID)
```

[49]:
```
                   mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
rmssd_within_z   -0.128  0.104  -0.329    0.067      0.001    0.001   14063.0
rmssd_between_z   0.157  0.117  -0.057    0.381      0.001    0.001   11667.0
1|groupID_sigma   0.200  0.137   0.000    0.439      0.002    0.001    2782.0
sigma             0.986  0.080   0.837    1.131      0.001    0.001    9784.0

                  ess_tail  r_hat
rmssd_within_z      6105.0    1.0
rmssd_between_z     6192.0    1.0
1|groupID_sigma     3039.0    1.0
sigma               5958.0    1.0
```

1.2.7 Performance

1.2.8 Model 14: PP_Z ~ realMeanHR_bpm_z + (1|groupID)

1. Statsmodels (LME4)

[50]:
```
model14_sm = sm.MixedLM.from_formula(
    "pp_z ~ realMean_HR_bpm_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result14_sm = model14_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 14 sm: PP_Z ~ realMean_HR_bpm_z + (1|groupID)")
print(result14_sm.summary())
```

```
Model 14 sm: PP_Z ~ realMean_HR_bpm_z + (1|groupID)
           Mixed Linear Model Regression Results
============================================================
Model:               MixedLM  Dependent Variable:  pp_z
No. Observations:    87       Method:              REML
No. Groups:          29       Scale:               0.7763
Min. group size:     3        Log-Likelihood:      -123.5054
Max. group size:     3        Converged:           Yes
Mean group size:     3.0
------------------------------------------------------------
                  Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept         -0.000    0.132 -0.000 1.000 -0.258  0.258
realMean_HR_bpm_z  0.041    0.107  0.388 0.698 -0.168  0.251
g Var              0.243    0.191
============================================================
```

2. Bambi (Robustness testing)

[51]:
```
model14_bmb = bmb.Model(
    "pp_z ~ realMean_HR_bpm_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model14 = model14_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)
```

```
print("Model 14 bmb: PP_Z ~ realMean_HR_bpm_z + (1|groupID)")
az.summary(idata_model14, var_names=["realMean_HR_bpm_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, realMean_HR_bpm_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 5 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 14 bmb: PP_Z ~ realMean_HR_bpm_z + (1|groupID)
```

[51]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	\
realMean_HR_bpm_z	0.036	0.109	-0.171	0.236	0.002	0.001	
1|groupID_sigma	0.472	0.183	0.115	0.834	0.005	0.003	
sigma	0.907	0.088	0.747	1.071	0.002	0.001	

	ess_bulk	ess_tail	r_hat
realMean_HR_bpm_z	5166.0	4753.0	1.0
1|groupID_sigma	1629.0	2249.0	1.0
sigma	2670.0	3436.0	1.0

Within/Between Model 14: PP_Z ~ hr_between_z + hr_within_z + (1|groupID)

1. Statsmodels (LME4)

[52]:
```
model14_within_sm = sm.MixedLM.from_formula(
    "pp_z ~ hr_within_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result14_within_sm = model14_within_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 14 within sm: PP_Z ~ hr_within_z + (1|groupID)")
print(result14_within_sm.summary())
```

```
Model 14 within sm: PP_Z ~ hr_within_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:            MixedLM Dependent Variable: pp_z
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              0.7695
Min. group size:  3       Log-Likelihood:     -122.9537
```

```
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
------------------------------------------------------
             Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------
Intercept    -0.000    0.130 -0.000 1.000 -0.254  0.254
hr_within_z   0.113    0.095  1.198 0.231 -0.072  0.299
g Var         0.232    0.182
======================================================
```

[53]:
```
model14_between_sm = sm.MixedLM.from_formula(
    "pp_z ~ hr_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result14_between_sm = model14_between_sm.fit(reml=True, method="nm",␣
 ↪maxiter=5000)
print("Model 14 between sm: PP_Z ~ hr_between_z + (1|groupID)")
print(result14_between_sm.summary())
```

```
Model 14 between sm: PP_Z ~ hr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
======================================================
Model:            MixedLM Dependent Variable: pp_z
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              0.7753
Min. group size:  3       Log-Likelihood:     -122.1915
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
------------------------------------------------------
             Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------
Intercept    -0.000    0.127 -0.000 1.000 -0.248  0.248
hr_between_z -0.197    0.127 -1.548 0.122 -0.447  0.052
g Var         0.207    0.174
======================================================
```

[54]:
```
model14_within_between_sm = sm.MixedLM.from_formula(
    "pp_z ~ hr_within_z + hr_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result14_within_between_sm = model14_within_between_sm.fit(reml=True,␣
 ↪method="nm", maxiter=5000)
```

```
print("Model 14 within-between sm: PP_Z ~ hr_within_z + hr_between_z +␣
 ↪(1|groupID)")
print(result14_within_between_sm.summary())
```

```
Model 14 within-between sm: PP_Z ~ hr_within_z + hr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: pp_z
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              0.7695
Min. group size:   3       Log-Likelihood:     -122.9138
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
             Coef.  Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept    -0.000    0.127 -0.000 1.000 -0.248  0.248
hr_within_z   0.113    0.095  1.198 0.231 -0.072  0.299
hr_between_z -0.197    0.127 -1.548 0.122 -0.447  0.052
g Var         0.209    0.175
=========================================================
```

2. Bambi (Robustness testing)

[55]:
```
model14_within_bmb  = bmb.Model(
    "pp_z ~ hr_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model14_within_bmb = model14_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 14 within bmb: PP_Z ~ hr_within_z + (1|groupID)")
az.summary(idata_model14_within_bmb, var_names=["hr_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, hr_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.

Model 14 within bmb: PP_Z ~ hr_within_z + (1|groupID)
```

[55]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
hr_within_z	0.114	0.098	-0.078	0.292	0.001	0.001	8723.0	5489.0	1.0
1|groupID_sigma	0.467	0.171	0.135	0.810	0.004	0.002	1749.0	2481.0	1.0
sigma	0.900	0.088	0.738	1.063	0.002	0.001	2716.0	3239.0	1.0

[56]:
```
model14_between_bmb  = bmb.Model(
    "pp_z ~ hr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model14_between_bmb = model14_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 14 between bmb: PP_Z ~ hr_between_z + (1|groupID)")
az.summary(idata_model14_between_bmb, var_names=["hr_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, hr_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.

Model 14 between bmb: PP_Z ~ hr_between_z + (1|groupID)
```

[56]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk
hr_between_z	-0.195	0.132	-0.457	0.049	0.002	0.002	5356.0
1|groupID_sigma	0.430	0.182	0.053	0.749	0.005	0.003	1344.0
sigma	0.905	0.085	0.753	1.068	0.002	0.001	2894.0

```
                  ess_tail  r_hat
hr_between_z        5331.0    1.0
1|groupID_sigma     1162.0    1.0
sigma               5258.0    1.0
```

[57]:
```
model14_within_between_bmb  = bmb.Model(
    "pp_z ~ hr_within_z + hr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model14_within_between_bmb = model14_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 14 within-between bmb: PP_Z ~ hr_within_z + hr_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model14_within_between_bmb, var_names=["hr_within_z",␣
 ↪"hr_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, hr_within_z, hr_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.
```

```
Model 14 within-between bmb: PP_Z ~ hr_within_z + hr_between_z + (1|groupID)
```

[57]:
```
                   mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
hr_within_z       0.113  0.097  -0.073    0.291      0.001    0.001    9624.0
hr_between_z     -0.194  0.131  -0.438    0.052      0.002    0.002    4866.0
1|groupID_sigma   0.429  0.183   0.037    0.730      0.005    0.003    1455.0
sigma             0.904  0.087   0.739    1.062      0.002    0.001    2840.0

                  ess_tail  r_hat
hr_within_z         6042.0    1.0
hr_between_z        5222.0    1.0
1|groupID_sigma     1684.0    1.0
```

```
sigma                4273.0     1.0
```

1.2.9 Model 15: PP_Z ~ realMeanRMSSD_bpm_z + (1|groupID)

1. Statsmodels (LME4)

[58]:
```
model15_sm = sm.MixedLM.from_formula(
    "pp_z ~ realRMSSD_ms_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result15_sm = model15_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 15 sm: PP_Z ~ realRMSSD_ms_z + (1|groupID)")
print(result15_sm.summary())
```

```
Model 15 sm: PP_Z ~ realRMSSD_ms_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: pp_z
No. Observations:   87      Method:             REML
No. Groups:         29      Scale:              0.7988
Min. group size:    3       Log-Likelihood:     -123.3808
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept      -0.000    0.127 -0.000 1.000 -0.249  0.249
realRMSSD_ms_z  0.068    0.110  0.619 0.536 -0.148  0.284
g Var           0.203    0.180
============================================================
```

2. Bambi (Robustness testing)

[59]:
```
model15_bmb = bmb.Model(
    "pp_z ~ realRMSSD_ms_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model15 = model15_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
```

```
)

print("Model 15 bmb: PP_Z ~ realRMSSD_ms_z + (1|groupID)")
az.summary(idata_model15, var_names=["realRMSSD_ms_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, realRMSSD_ms_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model 15 bmb: PP_Z ~ realRMSSD_ms_z + (1|groupID)
```

[59]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
realRMSSD_ms_z	0.076	0.111	-0.134	0.283	0.001	0.001	6300.0	
1|groupID_sigma	0.417	0.184	0.000	0.689	0.005	0.003	1309.0	
sigma	0.920	0.088	0.759	1.087	0.002	0.001	3230.0	

	ess_tail	r_hat
realRMSSD_ms_z	5135.0	1.0
1|groupID_sigma	1221.0	1.0
sigma	4985.0	1.0

Within/Between Model 15: PP_Z ~ rmssd_between_z + rmssd_within_z + (1|groupID)

1. Statsmodels (LME4)

[60]:
```
model15_within_sm = sm.MixedLM.from_formula(
    "pp_z ~ rmssd_within_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result15_within_sm = model15_within_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 15 within sm: PP_Z ~ rmssd_within_z + (1|groupID)")
print(result15_within_sm.summary())
```

```
Model 15 within sm: PP_Z ~ rmssd_within_z + (1|groupID)
          Mixed Linear Model Regression Results
=========================================================
Model:              MixedLM Dependent Variable: pp_z
No. Observations:   87      Method:             REML
No. Groups:         29      Scale:              0.7849
Min. group size:    3       Log-Likelihood:     -123.5157
```

```
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept      -0.000    0.130 -0.000 1.000 -0.254  0.254
rmssd_within_z -0.052    0.096 -0.543 0.587 -0.239  0.135
g Var           0.227    0.180
============================================================
```

[61]:
```
model15_between_sm = sm.MixedLM.from_formula(
    "pp_z ~ rmssd_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result15_between_sm = model15_between_sm.fit(reml=True, method="nm",␣
 ↪maxiter=5000)
print("Model 15 between sm: PP_Z ~ rmssd_between_z + (1|groupID)")
print(result15_between_sm.summary())
```

```
Model 15 between sm: PP_Z ~ rmssd_between_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:             MixedLM  Dependent Variable:  pp_z
No. Observations:  87       Method:              REML
No. Groups:        29       Scale:               0.7753
Min. group size:   3        Log-Likelihood:      -120.5983
Max. group size:   3        Converged:           Yes
Mean group size:   3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept      -0.000    0.119 -0.000 1.000 -0.234  0.234
rmssd_between_z 0.296    0.120  2.466 0.014  0.061  0.531
g Var           0.155    0.155
============================================================
```

[62]:
```
model15_within_between_sm = sm.MixedLM.from_formula(
    "pp_z ~ rmssd_within_z + rmssd_between_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result15_within_between_sm = model15_within_between_sm.fit(reml=True,␣
 ↪method="nm", maxiter=5000)
```

```
print("Model 15 within-between sm: PP_Z ~ rmssd_within_z + rmssd_between_z +␣
 ↪(1|groupID)")
print(result15_within_between_sm.summary())
```

```
Model 15 within-between sm: PP_Z ~ rmssd_within_z + rmssd_between_z +
(1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM  Dependent Variable:  pp_z
No. Observations:   87       Method:              REML
No. Groups:         29       Scale:               0.7849
Min. group size:    3        Log-Likelihood:      -121.8826
Max. group size:    3        Converged:           Yes
Mean group size:    3.0
------------------------------------------------------------
                Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       -0.000    0.119 -0.000 1.000 -0.234  0.234
rmssd_within_z  -0.052    0.096 -0.543 0.587 -0.239  0.135
rmssd_between_z  0.296    0.120  2.466 0.014  0.061  0.531
g Var            0.152    0.154
============================================================
```

2. Bambi (Robustness testing)

[63]:
```
model15_within_bmb  = bmb.Model(
    "pp_z ~ rmssd_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model15_within_bmb = model15_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 15 within bmb: PP_Z ~ rmssd_within_z + (1|groupID)")
az.summary(idata_model15_within_bmb, var_names=["rmssd_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, rmssd_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 5 divergences after tuning. Increase `target_accept` or
reparameterize.

Model 15 within bmb: PP_Z ~ rmssd_within_z + (1|groupID)
```

[63]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
rmssd_within_z	-0.053	0.098	-0.231	0.137	0.001	0.001	7839.0	4916.0	1.0
1|groupID_sigma	0.459	0.176	0.091	0.783	0.004	0.003	1873.0	2227.0	1.0
sigma	0.910	0.088	0.743	1.071	0.002	0.001	2997.0	3993.0	1.0

[64]:
```
model15_between_bmb  = bmb.Model(
    "pp_z ~ rmssd_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model15_between_bmb = model15_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 15 between bmb: PP_Z ~ rmssd_between_z + (1|groupID)")
az.summary(idata_model15_between_bmb, var_names=["rmssd_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, rmssd_between_z, 1|groupID_sigma, 1|groupID_offset]

Output()


Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.

Model 15 between bmb: PP_Z ~ rmssd_between_z + (1|groupID)
```

[64]:
```
                    mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
rmssd_between_z    0.295  0.124   0.053    0.518      0.002    0.002    5046.0
1|groupID_sigma    0.372  0.173   0.001    0.635      0.004    0.002    1606.0
sigma              0.902  0.083   0.750    1.058      0.001    0.001    3309.0

                   ess_tail  r_hat
rmssd_between_z      4751.0    1.0
1|groupID_sigma      1867.0    1.0
sigma                5638.0    1.0
```

[65]:
```
model15_within_between_bmb  = bmb.Model(
    "pp_z ~ rmssd_within_z + rmssd_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model15_within_between_bmb = model15_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 15 within-between bmb: PP_Z ~ rmssd_within_z + rmssd_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model15_within_between_bmb, var_names=["rmssd_within_z",␣
 ↪"rmssd_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, rmssd_within_z, rmssd_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.
```

```
Model 15 within-between bmb: PP_Z ~ rmssd_within_z + rmssd_between_z +
(1|groupID)
```

[65]:
```
                    mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
rmssd_within_z    -0.052  0.098  -0.240    0.125      0.001    0.001    9529.0
rmssd_between_z    0.296  0.123   0.065    0.527      0.002    0.001    6557.0
1|groupID_sigma    0.364  0.174   0.007    0.637      0.004    0.002    1634.0
sigma              0.908  0.083   0.752    1.062      0.001    0.001    3716.0
```

```
                  ess_tail  r_hat
rmssd_within_z      5990.0    1.0
rmssd_between_z     5693.0    1.0
1|groupID_sigma     2311.0    1.0
sigma               5206.0    1.0
```

1.2.10 Model 16: PP_Z ~ me + (1|groupID)

1. Statsmodels (LME4)

[66]:
```
model16_sm = sm.MixedLM.from_formula(
    "pp_z ~ me_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result16_sm = model16_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 16 sm: PP_Z ~ me_z + (1|groupID)")
print(result16_sm.summary())
```

```
Model 16 sm: PP_Z ~ me_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: pp_z
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              0.7184
Min. group size:   3       Log-Likelihood:     -117.5264
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
           Coef.  Std.Err.    z    P>|z|  [0.025  0.975]
---------------------------------------------------------
Intercept  0.000     0.115  0.000  1.000  -0.225   0.225
me_z       0.354     0.100  3.555  0.000   0.159   0.550
g Var      0.143     0.151
=========================================================
```

2. Bambi (Robustness testing)

[67]:
```
model16_bmb = bmb.Model(
    "pp_z ~ me_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model16 = model16_bmb.fit(
```

```
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 16 bmb: PP_Z ~ me_z + (1|groupID)")
az.summary(idata_model16, var_names=["me_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, me_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 19 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 16 bmb: PP_Z ~ me_z + (1|groupID)
```

[67]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
me_z	0.357	0.101	0.169	0.550	0.001	0.001	6724.0	
1|groupID_sigma	0.348	0.168	0.001	0.603	0.004	0.002	1690.0	
sigma	0.871	0.082	0.721	1.025	0.001	0.001	3041.0	

	ess_tail	r_hat
me_z	4630.0	1.0
1|groupID_sigma	2609.0	1.0
sigma	2848.0	1.0

1.2.11 Model 17: PP_Z ~ td + (1|groupID)

1. Statsmodels (LME4)

[68]:
```
model17_sm = sm.MixedLM.from_formula(
    "pp_z ~ td_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
result17_sm = model17_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model 17 sm: PP_Z ~ td_z + (1|groupID)")
print(result17_sm.summary())
```

```
Model 17 sm: PP_Z ~ td_z + (1|groupID)
         Mixed Linear Model Regression Results
```

```
=========================================================
Model:             MixedLM Dependent Variable: pp_z
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              0.7907
Min. group size:   3       Log-Likelihood:     -121.0131
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
             Coef.  Std.Err.    z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept    -0.000    0.118 -0.000 1.000 -0.231  0.231
td_z         -0.245    0.108 -2.268 0.023 -0.456 -0.033
g Var         0.141    0.156
=========================================================
```

2. Bambi (Robustness testing)

[69]:
```
model17_bmb = bmb.Model(
    "pp_z ~ td_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model17 = model17_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 17 bmb: PP_Z ~ td_z + (1|groupID)")
az.summary(idata_model17, var_names=["td_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, td_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 4 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 17 bmb: PP_Z ~ td_z + (1|groupID)
```

[69]:
```
                    mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
td_z              -0.248  0.108  -0.458   -0.052      0.001    0.001    6823.0
1|groupID_sigma    0.350  0.175   0.000    0.621      0.004    0.002    1633.0
sigma              0.910  0.084   0.758    1.070      0.001    0.001    3531.0

                  ess_tail  r_hat
td_z                5237.0    1.0
1|groupID_sigma     2590.0    1.0
sigma               3572.0    1.0
```

1.2.12 Model 18: uncertScore ~ me + (1|groupID)

1. Statsmodels (LME4)

[76]:
```
# Model 18: team-level OLS (no groups, no random effect)
result18_ols = smf.ols("uncertScore_z ~ me_between_z", data=team_model).fit()
print(result18_ols.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          uncertScore_z   R-squared:                       0.010
Model:                            OLS   Adj. R-squared:                 -0.027
Method:                 Least Squares   F-statistic:                    0.2640
Date:                Fri, 25 Sep 2026   Prob (F-statistic):              0.612
Time:                        16:31:54   Log-Likelihood:                -40.841
No. Observations:                  29   AIC:                             85.68
Df Residuals:                      27   BIC:                             88.42
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept  -4.025e-16      0.190  -2.11e-15      1.000      -0.391       0.391
me_between_z    -0.0984      0.192     -0.514      0.612      -0.491       0.295
==============================================================================
Omnibus:                        6.652   Durbin-Watson:                   1.878
Prob(Omnibus):                  0.036   Jarque-Bera (JB):                5.014
Skew:                          -0.743   Prob(JB):                       0.0815
Kurtosis:                       4.394   Cond. No.                         1.01
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
```

2. Bambi (Robustness testing)

[ ]:
```
model18_bmb = bmb.Model(
    "uncertScore_z ~ me_between_z",
    data=team_model,
    family="gaussian"
)

idata_model18 = model18_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)


print("Model 18 bmb: uncertScore_z ~ me_between_z")
az.summary(idata_model18, var_names=["me_between_z", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, me_between_z]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
```

```
Model 18 bmb: uncertScore_z ~ me_between_z
```

[ ]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
me_between_z	-0.098	0.199	-0.454	0.307	0.002	0.002	9004.0	
sigma	1.051	0.146	0.796	1.329	0.002	0.002	7564.0	

	ess_tail	r_hat
me_between_z	5645.0	1.0
sigma	4594.0	1.0

1.2.13 Model 19: uncertScore_z ~ td + (1|groupID)

1. Statsmodels (LME4)

[74]:
```
result19_ols = smf.ols("uncertScore_z ~ td_between_z", data=team_model).fit()
print(result19_ols.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          uncertScore_z   R-squared:                       0.003
Model:                            OLS   Adj. R-squared:                 -0.034
Method:                 Least Squares   F-statistic:                    0.07776
```

```
Date:                Fri, 25 Sep 2026   Prob (F-statistic):              0.782
Time:                        16:31:44   Log-Likelihood:                -40.940
No. Observations:                  29   AIC:                             85.88
Df Residuals:                      27   BIC:                             88.61
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept  -4.025e-16      0.191  -2.11e-15      1.000      -0.392       0.392
td_between_z    -0.0536      0.192     -0.279      0.782      -0.448       0.341
==============================================================================
Omnibus:                        5.127   Durbin-Watson:                   1.790
Prob(Omnibus):                  0.077   Jarque-Bera (JB):                3.384
Skew:                          -0.711   Prob(JB):                        0.184
Kurtosis:                       3.881   Cond. No.                         1.01
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
```

2. Bambi (Robustness testing)

[78]:
```
model19_bmb = bmb.Model(
    "uncertScore_z ~ td_between_z",
    data=team_model,
    family="gaussian"
)

idata_model19 = model19_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)


print("Model 19 bmb: uncertScore_z ~ td_between_z")
az.summary(idata_model19, var_names=["td_between_z", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, td_between_z]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.

Model 19 bmb: uncertScore_z ~ td_between_z
```

[78]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
td_between_z	-0.053	0.201	-0.454	0.300	0.002	0.002	9651.0	
sigma	1.056	0.147	0.786	1.316	0.002	0.002	7523.0	

	ess_tail	r_hat
td_between_z	6004.0	1.0
sigma	5223.0	1.0

1.2.14 EXTRA

1.2.15 Model 1: me_z ~ td_z + (1|groupID)

1. Statsmodels (LME4)

[ ]:
```
modelEXTRA1_sm = sm.MixedLM.from_formula(
    "me_z ~ td_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
resultEXTRA1_sm = modelEXTRA1_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model EXTRA1 sm: ME_Z ~ TD_Z + (1|groupID)")
print(resultEXTRA1_sm.summary())
```

```
Model EXTRA1 sm: ME_Z ~ TD_Z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: me_z
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              1.0108
Min. group size:   3       Log-Likelihood:     -125.5256
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
             Coef.  Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept    -0.000    0.108 -0.000 1.000 -0.211  0.211
td_z          0.031    0.111  0.281 0.778 -0.186  0.249
g Var         0.000    0.117
=========================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:2237: ConvergenceWarning:
The MLE may be on the boundary of the parameter space.
```

```
warnings.warn(msg, ConvergenceWarning)
```

2. Bambi (Robustness testing)

[ ]:
```
modelENTRA1_bmb = bmb.Model(
    "me_z ~ td_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_modelEXTRA1 = modelENTRA1_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)




print("Model EXTRA1 bmb: ME_Z ~ TD_Z + (1|groupID)")
az.summary(idata_modelEXTRA1, var_names=["td_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, td_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 7 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model EXTRA1 bmb: ME_Z ~ TD_Z + (1|groupID)
```

[ ]:
```
                  mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
td_z             0.040  0.112  -0.172    0.247      0.001    0.002    9318.0
1|groupID_sigma  0.209  0.144   0.000    0.458      0.002    0.002    3151.0
sigma            0.997  0.080   0.854    1.155      0.001    0.001    6896.0

                 ess_tail  r_hat
td_z               4577.0    1.0
1|groupID_sigma    3941.0    1.0
sigma              4470.0    1.0
```

1.2.16 Model 2: td_z ~ me_z + (1|groupID)

1. Statsmodels (LME4)

[ ]:
```
modelEXTRA2_sm = sm.MixedLM.from_formula(
    "td_z ~ me_z",
    groups="g",
    re_formula="1",
    data=dat_model
)
resultEXTRA2_sm = modelEXTRA2_sm.fit(reml=True, method="nm", maxiter=5000)
print("Model EXTRA2 sm: TD_Z ~ ME_Z + (1|groupID)")
print(resultEXTRA2_sm.summary())
```

```
Model EXTRA2 sm: TD_Z ~ ME_Z + (1|groupID)
         Mixed Linear Model Regression Results
============================================================
Model:             MixedLM Dependent Variable: td_z
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              0.9566
Min. group size:   3       Log-Likelihood:     -125.4061
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
------------------------------------------------------------
            Coef.  Std.Err.    z    P>|z|  [0.025  0.975]
------------------------------------------------------------
Intercept   0.000     0.114  0.000  1.000  -0.223   0.223
me_z        0.041     0.110  0.377  0.706  -0.174   0.257
g Var       0.056     0.127
============================================================
```

2. Bambi (Robustness testing)

[ ]:
```
modelENTRA2_bmb = bmb.Model(
    "td_z ~ me_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_modelEXTRA2 = modelENTRA2_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model EXTRA2 bmb: TD_Z ~ ME_Z + (1|groupID)")
```

```
az.summary(idata_modelEXTRA2, var_names=["me_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, me_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 2 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model EXTRA2 bmb: TD_Z ~ ME_Z + (1|groupID)
```

[ ]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
me_z	0.046	0.110	-0.159	0.254	0.001	0.001	11942.0	
1|groupID_sigma	0.263	0.162	0.000	0.534	0.003	0.002	2233.0	
sigma	0.982	0.084	0.832	1.147	0.001	0.001	6268.0	

	ess_tail	r_hat
me_z	5519.0	1.0
1|groupID_sigma	3355.0	1.0
sigma	4165.0	1.0